

Agentic AI Case Study

Problem Statement

Build a Student Enrollment Assistant Agent

You are building a conversational AI agent for a university's admissions office. The agent helps prospective students get answers about programs, deadlines, and application status. It must use tools (not just LLM knowledge) to answer questions.

Requirements

1. Define three tools the agent can call:

Tool	Input	Returns
<code>get_program_info</code>	<code>program_name</code> : <code>str</code>	Dict with program name, duration, tuition, and prerequisites
<code>check_application_status</code>	<code>applicant_id</code> : <code>str</code>	Dict with applicant name, program applied to, status (e.g., "Under Review", "Accepted", "Documents Pending"), and next step
<code>get_deadlines</code>	<code>program_name</code> : <code>str</code>	Dict with application deadline, document submission deadline, and decision notification date

The tools should use hardcoded/mock data (no real database needed). Provide at least 3 programs and 3 applicants in the mock data.

2. Build the agent loop:

- The agent receives a user message, reasons about which tool(s) to call, executes them, and formulates a natural language response.

- The agent must support multi-turn conversation — it should remember prior context within a session (e.g., if a student already mentioned their applicant ID, the agent shouldn't ask again).
- If the user's question cannot be answered by the available tools, the agent should gracefully escalate — respond with something like: *"I'd recommend speaking with an enrollment counselor for that. Would you like me to connect you?"*

3. Demonstrate with a test conversation:

Run the agent through the following 5-turn conversation and include the full input/output log:

text

Turn 1: "Hi, what programs do you offer in computer science?"

Turn 2: "What's the application deadline for that?"

Turn 3: "I already applied. My ID is APP-1042. What's my status?"

Turn 4: "Can I get a fee waiver?"

Turn 5: "What documents do I still need to submit?"